

SIMATS ENGINEERING

Swetha.B
(192412215)

CO1-AT1

Short answer questions

ITA0606-
Machine learning

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11. Bayes' Theorem – Oil Bottles

Given

Shop X:

- Pure = 20
- Adulterated = 30
- Total = 50

Shop Y:

- Pure = 40
- Adulterated = 20
- Total = 60

Shop Y is twice as likely as Shop X.

$$\begin{aligned}P(X) &= \frac{1}{3}, & P(Y) &= \frac{2}{3} \\P(A|X) &= \frac{30}{50} = 0.6 \\P(A|Y) &= \frac{20}{60} = \frac{1}{3}\end{aligned}$$

where A = Adulterated oil

Using Bayes' Theorem

$$P(X|A) = \frac{P(A|X)P(X)}{P(A|X)P(X) + P(A|Y)P(Y)}$$

Substituting values:

$$\begin{aligned}P(X|A) &= \frac{0.6 \times \frac{1}{3}}{0.6 \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3}} \\&= \frac{0.2}{0.2 + 0.2222} \\&= \frac{0.2}{0.4222} \\&= 0.474\end{aligned}$$

$$P(X|A)=0.474$$

12. Bayes' Theorem – Warehouse Defective Items

Given

Section A:

- Defective = 15
- Non-defective = 35
- Total = 50

Section B:

- Defective = 25
- Non-defective = 25
- Total = 50

Section A is three times as likely as Section B.

$$P(A) = \frac{3}{4}$$

$$P(B) = \frac{1}{4}$$

$$P(D|A) = \frac{15}{50} = 0.3$$

$$P(D|B) = \frac{25}{50} = 0.5$$

Using Bayes' Theorem

$$\begin{aligned} P(A|D) &= \frac{P(D|A)P(A)}{P(D|A)P(A) + P(D|B)P(B)} \\ &= \frac{0.3 \times 0.75}{0.3 \times 0.75 + 0.5 \times 0.25} \\ &= \frac{0.225}{0.225 + 0.125} \\ &= \frac{0.225}{0.35} \\ &= 0.643 \end{aligned}$$

Conclusion

The probability that the defective item came from Section A is 64.3%

13. Accuracy, Precision, Recall and Specificity

Given

$$TP = 120$$

$$TN = 150$$

Each class contains 1000 samples.

Total samples:

$$1000 + 1000 = 2000$$

$$\text{Test data} = 20\%$$

$$0.2 \times 2000 = 400$$

$$\text{Actual Positive} = 200$$

$$\text{Actual Negative} = 200$$

Find FP and FN

$$FN = 200 - TP$$

$$FN = 200 - 120 = 80$$

$$FP = 200 - TN$$

$$FP = 200 - 150 = 50$$

Accuracy

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ &= \frac{120 + 150}{120 + 150 + 50 + 80} \\ &= \frac{270}{400} \\ &= 67.5\% \end{aligned}$$

Precision

$$\begin{aligned} Precision &= \frac{TP}{TP + FP} \\ &= \frac{120}{120 + 50} \\ &= \frac{120}{170} \\ &= 70.59\% \end{aligned}$$

Recall (Sensitivity)

$$\begin{aligned} Recall &= \frac{TP}{TP + FN} \\ &= \frac{120}{120 + 80} \\ &= \frac{120}{200} \\ &= 60\% \end{aligned}$$

14. Medical Image Classification Metrics

Given

TP = 180

TN = 160

Each class contains 1200 samples.

Total samples:

1200+1200=2400

Test data = 25%

0.25×2400=600

Actual Positive = 300

Actual Negative = 300

Find FP and FN

$$FN = 300 - 180 = 120$$

$$FP = 300 - 160 = 140$$

Accuracy

$$\begin{aligned} Accuracy &= \frac{180 + 160}{180 + 160 + 140 + 120} \\ &= \frac{340}{600} \\ &= 56.67\% \end{aligned}$$

Precision

$$\begin{aligned} Precision &= \frac{180}{180 + 140} \\ &= \frac{180}{320} \\ &= 56.25\% \end{aligned}$$

Sensitivity (Recall)

$$\begin{aligned} Sensitivity &= \frac{180}{180 + 120} \\ &= \frac{180}{300} \\ &= 60\% \end{aligned}$$

Specificity

$$\begin{aligned} Specificity &= \frac{160}{160 + 140} \\ &= \frac{160}{300} \\ &= 53.33\% \end{aligned}$$

1. Probability Using Bayes' Theorem

Given

Number of balls in each box

Box A

- Red = 6
- Blue = 4
- Green = 2

Total = 12

$$P(BL|A) = \frac{4}{12} = \frac{1}{3}$$

Box B

- Red = 5
- Blue = 7
- Green = 3

Total = 15

$$P(BL|B) = \frac{7}{15}$$

Box C

- Red = 8
- Blue = 2
- Green = 5

Total = 15

$$P(BL|C) = \frac{2}{15}$$

Selection Probabilities

Box B is twice as likely to be chosen as Box A and Box C.

Let

$$P(A) = x, \quad P(C) = x, \quad P(B) = 2x$$

Since total probability = 1,

$$x + 2x + x = 1$$

$$4x = 1$$

$$x = \frac{1}{4}$$

Therefore,

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{4}$$

Using Bayes' Theorem

We need:

$$P(C|BL) = \frac{P(BL|C)P(C)}{P(BL|A)P(A) + P(BL|B)P(B) + P(BL|C)P(C)}$$

Substituting values:

$$\begin{aligned} P(C|BL) &= \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{1}{3} \times \frac{1}{4} + \frac{7}{15} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}} \\ &= \frac{\frac{1}{30}}{\frac{1}{12} + \frac{7}{30} + \frac{1}{30}} \end{aligned}$$

Taking LCM = 60,

$$\begin{aligned} &= \frac{\frac{2}{60}}{\frac{5}{60} + \frac{14}{60} + \frac{2}{60}} \\ &= \frac{2}{21} \\ &= 0.0952 \end{aligned}$$